

POLOXAMERS INFLUENCE THE DISSOLUTION AND STABILITY OF LIPID MATRICES OF PRECIROL ATO 5™

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I - PURPOSE

When used in relatively high quantities, Precirol ATO 5™ imparts sustained release properties to solid oral dosage forms, by forming a lipid matrix. To control or adjust the drug release kinetics from such lipid matrix however, one must often resort to complementary ingredients or techniques. Addition of hydrophilic components to Precirol ATO 5™ for example, has shown interesting improvements in drug release adjustment or solubilization of the drug in the matrix system by spray-chilling [1] or tableting [2]. The aim of this study was to investigate the influence of two hydrophilic polymers, Poloxamers 188 & 407 on the dissolution performance and stability of Precirol ATO 5™ lipid-matrices obtained by capsule molding.

II - METHODS

• Formulation of sustained release matrices in capsules

Poloxamers 188 or 407 were stirred-mixed into molten Precirol ATO 5™ (glyceryl distearate). Theophylline was dispersed into the mixtures before pouring into capsules. Formulations with varying percentages of Precirol ATO 5™ and Poloxamer were defined according to a factorial design of experiments (Table 1).

Table 1: Compositions of sustained-release formulations (*performed in triplicate).

Formulations	Theophylline (% w/w)	Precirol ATO 5™ (% w/w)	Poloxamer 407 (% w/w)	Poloxamer 188 (% w/w)
F1	14.7	60.3	25	-
F2	14.7	60.3	-	25
F3*	14.7	70.3	-	15
F4	14.7	80.3	5	-
F5	14.7	80.3	-	5

• Physical characterization of matrices

Interactions between components were assessed by DSC (Pyris Diamond, Perkin Elmer, USA) and hot-stage microscopy (Leitz Wetzlar, Germany equipped with a heating unit FP2HT, Mettler Toledo, France).

• Dissolution performance of matrices

The influence of poloxamers on the dissolution performance of matrices was examined by dissolution testing (AT7, Sotax, Switzerland). The mechanism of release was assessed by rheological (Contraves Rheomat 115, Contraves Rheoanalyser, Switzerland), water-uptake and erosion testing.

III - RESULTS

The main interaction observed by thermal analysis is between Precirol ATO 5™ and poloxamers. Addition of poloxamers in a solid matrix of Precirol ATO 5™ increased the quantity of theophylline released because of the swelling of the matrices and the formation of pores (Fig. 1).

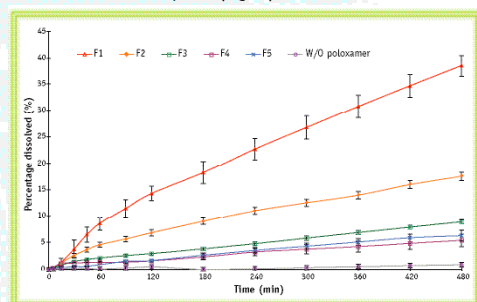


Fig. 1: Dissolution profiles of theophylline within formulations composed of Precirol ATO 5™ with or without poloxamers.

Increasing the poloxamers percentage in the formula also positively influenced the drug release and the dissolution efficiency of the formulations. The matrices underwent a gradual swelling that increased with the quantity of poloxamers. The grade of poloxamers also influenced the dissolution; formulations with poloxamers 407 presented a swelling ratio greater than those with poloxamers 188 and led to a more porous system hence a greater drug release (Fig. 2).

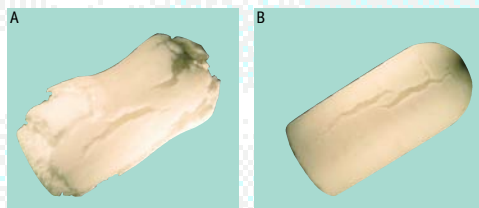


Fig. 2: Formulations F1 (A) and F2 (B) after 8 hours of dissolution and drying onto silica gel at room temperature.

Furthermore the apparent viscosity and dissolution efficiency (DE) of the formulations increased with the poloxamers' addition (Table 2).

Table 2: HLB value, dissolution efficiency and apparent viscosity at 60°C of the five formulations.

Formulations	HLB	Dissolution efficiency %	Apparent viscosity at 60°C (mPa. s)
F1	7.9	22	506
F2	9.9	10	315
F3	6.7	5	180
F4	3.2	3	111
F5	3.6	3	113

Theophylline release from the five formulations was correlated to their apparent viscosity:

$$DE = 1.4743 + 0.0074\eta + 0.0001\eta^2 \quad R^2 = 0.9998$$

These profiles could fit the Peppas' model: $F = k.t^n$ for drug release (Table 3).

Table 3: Non-linear regression of the dissolution profiles with Peppas' model.

Formulations	k	n	R ²
F1	0.3499	0.7609	0.9971
F2	0.2496	0.6869	0.9975
F3	0.0826	0.7499	0.9919
F4	0.0287	0.8475	0.9968
F5	0.0259	0.8918	0.9985

Dissolution profiles after 12 months at 25°C/60% RH and 6 months at 40°C/75% RH illustrated in Fig. 3 are promising, indicative of formulation stability over time.

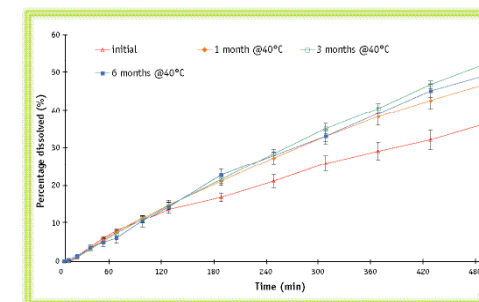
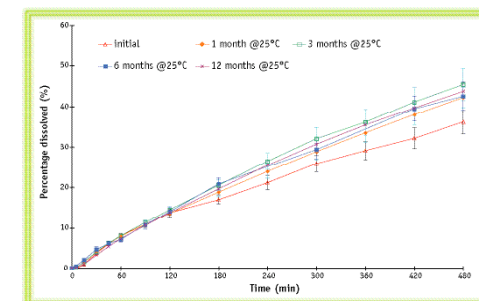


Fig. 3: Dissolution profiles of formulation F1 stored at 25°C/60% RH and 40°C/75% RH.

IV - CONCLUSION

The development of a theophylline controlled-release formulation through a mixture of Precirol ATO 5™ and poloxamers 407 showed many advantages compared to classical lipid matrices such as an increase in the drug release and a stabilization of the dissolution profile over time.

V - REFERENCE LIST

1. Savolainen M, Herder J, Khoo C et al. Evaluation of polar lipid-hydrophilic polymer microparticles. *Int J Pharm* 2003; 262 (1-2): 47-62.
2. Parob PV, Oh CK, Ritschel WA. Sustained release from Precirol™ (glycerol palmito-stearate) matrix. Effect of mannitol and hydroxypropylmethylcellulose on the release of theophylline. *Drug Dev Ind Pharm* 1986; 12(8-9): 1309-1327.